

Question Number	Answer	Additional Guidance	Mark
3a	There is no (significant) difference between the (mean) mass of leaves eaten of variety A and variety B	Candidates can express this in different ways	1

3b	• suitable table format with correct column headings and units (1) • all data correctly entered (1) • means correctly calculated (1)	<table><tr><th colspan="2">Mass of leaves eaten (in 24 hours) / g</th></tr><tr><th>(variety) A</th><th>(variety) B</th></tr><tr><td>17.3</td><td>17.8</td></tr><tr><td>17.2</td><td>17.9</td></tr><tr><td>17.5</td><td>17.7</td></tr><tr><td>17.0</td><td>17.6</td></tr><tr><td>16.7</td><td>17.8</td></tr><tr><td>16.9</td><td>17.4</td></tr><tr><td>Mean 17.1</td><td>Mean 17.7</td></tr></table>	Mass of leaves eaten (in 24 hours) / g		(variety) A	(variety) B	17.3	17.8	17.2	17.9	17.5	17.7	17.0	17.6	16.7	17.8	16.9	17.4	Mean 17.1	Mean 17.7	3
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3c	- bar graph with linear scale and axes labelled, with units (1) - means plotted correctly (1) - range bars plotted correctly (1)	Mean mass of leaves (eaten) /g A B ALLOW ECF from 3bi for plotting incorrect means 16.7 to 17.5 17.4 to 17.9	3

Question Number	Answer	Additional Guidance	Mark
3di	- correct substitution of given $(S_A)^2$ and $(S_B)^2$ (1) - correct answer (1)	0.084 and 0.032 $t = 4.3 / 4.32 / 4.315 / 4.3152$ Correct answer gains 2 marks Ignore minus signs Allow ECF using incorrect means Denominator is 0.139	2

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3d ii	- the calculated value of t (4.3152) is more than the critical value 2.23 (1) - therefore reject the null hypothesis, so there is a difference in the mass (of leaves) eaten between variety A and variety B (1)	Do not allow -ve values / $t < \text{critical value}$ if t value not given Ignore $p = 0.01$ Allow ECF for MP2	2

Question Number	Answer	Additional Guidance	Mark
3e	An answer that includes two of the following points: - use different temperatures (1) - use different {species / ages / stages of locust} (1) - leave for longer than 24 hrs (1) - use both varieties at once so locust can select diet (1)	Ignore repeat the expt / use more locusts Allow different humidities Ignore types	2

Question Number	Answer	Additional Guidance	Mark
3f	An answer that includes one of the following points: • locusts {eat more at higher temperatures / have a faster metabolism / respire faster so they grow faster} (1) • {locusts complete their life cycle faster / more locusts produced} (1)	Allow environmental answers e.g. increase in drought / temperature too hot for plant growth / more extreme weather such as severe storms	1

(Total for question 3 = 14 marks)

